Logarithmic algebraic geometry: affine toric varieties and log structures

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1 Rational polyhedral cones and toric monoids

Let $N = \mathbb{Z}^{\oplus r}$ for $r \geq 0$, and let $N_{\mathbb{R}} = \mathbb{R}^{\oplus r} \cong N \otimes_{\mathbb{Z}} \mathbb{R}$. A *rational polyhedral cone* is a closed subset $\sigma \subset N_{\mathbb{R}}$ of the form

$$\sigma = \mathbb{R}_{\geq 0} \cdot \{n_1, \ldots, n_k\} = \{\lambda_1 n_1 + \ldots + \lambda_k n_k \mid \lambda_i \geq 0\} \subset N_{\mathbb{R}}$$

for fixed vectors $n_1, \ldots, n_k \in N$. A rational polyhedral cone is called *strongly convex* if $n = 0$ is the only $n \in \sigma$ with $-n \in \sigma$.

Example 1.1. (1) $\sigma_a = [0, \infty) \subset \mathbb{R}$ is a strongly convex rational polyhedral cone.

(2) $\sigma_b = \mathbb{R} \subset \mathbb{R}$ is a rational polyhedral cone which is not strongly convex.

(3) $\sigma_c = \mathbb{R}_{\geq 0} \cdot (0,1) + \mathbb{R}_{\geq 0} \cdot (-1,2) \subset \mathbb{R}^{\oplus 2}$ is a strongly convex rational polyhedral cone.

(4) $\sigma_d = \mathbb{R}_{\geq 0} \cdot (1,0) + \mathbb{R}_{\geq 0} \cdot (0,1) \subset \mathbb{R}^2$ is a strongly convex rational polyhedral cone.

(5) $\sigma_e = \mathbb{R}_{\geq 0} \cdot (1,1) \subset \mathbb{R}^2$ is a strongly convex rational polyhedral cone. It is not full-dimensional.

(6) $\sigma_f = \mathbb{R}_{\geq 0} \cdot (1,0) + \mathbb{R} \cdot (0,1)$ is a rational polyhedral cone which is not strongly convex. $\lozenge$

Let $M = \mathrm{Hom}_{\mathbb{Z}}(N, \mathbb{Z})$ and $M_{\mathbb{R}} = \mathrm{Hom}_{\mathbb{R}}(N_{\mathbb{R}}, \mathbb{R}) \cong M \otimes_{\mathbb{Z}} \mathbb{R}$. For a rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$, we define its *dual cone* as

$$\sigma^\vee = \{m \in M_{\mathbb{R}} \mid \forall\, n \in \sigma : \langle m, n \rangle \geq 0\} \subset M_{\mathbb{R}}.$$

Fact 1.2. *We have $(\sigma^\vee)^\vee = \sigma$ under the identification $N_{\mathbb{R}} \cong \mathrm{Hom}_{\mathbb{R}}(M_{\mathbb{R}}, \mathbb{R})$.*

Fact 1.3. *The dual cone $\sigma^\vee \subset M_{\mathbb{R}}$ is a rational polyhedral cone, i.e., there are $m_1, \ldots, m_\ell \in M$ with $\sigma^\vee = \mathbb{R}_{\geq 0} \cdot \{m_1, \ldots, m_\ell\}$.*

Corollary 1.4. *There is a finite sequence of elements $m_1, \ldots, m_\ell \in M$ such that $\sigma = \bigcap_{j=1}^{\ell} \{\langle m_j, n \rangle \geq 0\}$.*

Now, for every rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$, we define its associated *monoid* as

$$P_\sigma \coloneqq \sigma^\vee \cap M \subset M.$$

Definition 1.5. A *commutative monoid* is a set P together with a binary operation $P \times P \to P$ which is associative, commutative, and admits a neutral element $0 \in P$. A *monoid homomorphism* is a set-theoretic map $\theta\colon Q \to P$ such that $\theta(0) = 0$ and $\theta(q_1 + q_2) = \theta(q_1) + \theta(q_2)$. We write CMon for the category of commutative monoids. ♦

We usually simply say "monoid" for a commutative monoid, with commutativity understood.

Example 1.6. (1) Let A be an Abelian group. Then, A is a commutative monoid. This defines a fully faithful functor $\mathrm{Ab} \to \mathrm{CMon}$.

(2) $\mathbb{N} = \{0, 1, 2, \ldots\}$ is a commutative monoid.

(3) $\mathbb{N}^{\oplus s}$ is a commutative monoid. ◊

Lemma 1.7. *Let $\sigma \subset N_{\mathbb{R}}$ be a rational polyhedral cone. Then, $P_\sigma = \sigma^\vee \cap M$ is a commutative monoid.*

Proof. It is sufficient to show that $P_\sigma \subseteq M$ contains the neutral element and is closed under the addition in M. We clearly have $0 \in P_\sigma$. Moreover, if $p_1, p_2 \in P_\sigma$, then $p_1 + p_2 \in M$, and $\langle p_1 + p_2, n\rangle = \langle p_1, n\rangle + \langle p_2, n\rangle \geq 0$ for all $n \in \sigma$. Thus, we have $p_1 + p_2 \in P_\sigma$. □

Remark 1.8. $\sigma \subset N_{\mathbb{R}}$, $\sigma^\vee \subset M_{\mathbb{R}}$, and $\sigma \cap N$ are commutative monoids as well. ◊

A monoid P is *finitely generated* if there is a surjective monoid homomorphism $\mathbb{N}^{\oplus s} \to P$.

Example 1.9. (1) Let $\sigma = \sigma_c = \mathbb{R}_{\geq 0} \cdot (1, 0) + \mathbb{R}_{\geq 0} \cdot (2, 1)$. Then, $P_2 \coloneqq P_\sigma$ is generated by $(-1, 2)$, $(1, 0)$, and $(0, 1)$.

(2) $(\mathbb{Z}, +)$ is finitely generated (by 1 and -1), but $(\mathbb{Z}, \cdot)$ is not finitely generated—it is generated by $0, -1$, and all prime numbers. ◊

Lemma 1.10 (Gordon's lemma)**.** *Let σ be a rational polyhedral cone. Then, P_σ is finitely generated.*

Proof. Let $m_1, \ldots, m_\ell \in M$ with $\sigma^\vee = \mathbb{R}_{\geq 0} \cdot m_1 + \ldots + \mathbb{R}_{\geq 0} \cdot m_\ell$. Let

$$K = \left\{ \sum_{i=1}^{\ell} t_i m_i \;\middle|\; t_i \in [0, 1] \right\}.$$

This is a compact subset of $M_{\mathbb{R}}$; in particular, $K \cap M$ is finite. Now, let $m \in P_\sigma$. Then, we can write $m = \sum_{i=1}^{\ell} \lambda_i u_i$ with $\lambda_i = k_i + t_i$ for $k_i \in \mathbb{N}$ and $t_i \in [0, 1]$. Thus, m is a sum of the elements $m_1, \ldots, m_\ell$ and some element in $K \cap M$. □

The forgetful functor $\mathrm{Ab} \to \mathrm{CMon}$ has right adjoint $(-)^\times\colon \mathrm{CMon} \to \mathrm{Ab}$. It is given by

$$P^\times = \{p \in P \mid \exists\, q \in P : p + q = 0\}.$$

We say that $P^\times$ is the *subgroup of invertible elements*. A monoid is called *sharp* if $P^\times = \{0\}$. In particular, a rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$ is strongly convex if and only if $\sigma^\times = \{0\}$. For cones in toric geometry, the terminology "strongly convex" is standard while for monoids more generally, and for the monoid P_σ in particular, the terminology "sharp" is more normal.

Example 1.11. (1) $\mathbb{N}^\times = \{0\}$

(2) $(\mathbb{Z},+)^\times = \mathbb{Z}$

(3) $(\mathbb{Z},\cdot)^\times = \{\pm 1\}$ ◊

Exercise 1.12. Show that $P \mapsto P^\times$ is indeed a right adjoint to Ab $\to$ CMon.

The forgetful functor Ab $\to$ CMon also has a left adjoint, the *Grothendieck group* functor

$$(-)^{\mathrm{gp}}\colon \ \mathrm{CMon} \to \mathrm{Ab}.$$

It is constructed as follows:

Construction 1.13. Let P be a monoid. On the set of pairs $P \times P$, we consider an equivalence relation given by $(p_1, d_1) \sim (p_2, d_2)$ if and only if there is some $q \in P$ with $p_1 + d_2 + q = p_2 + d_1 + q$. As a set, we set $P^{\mathrm{gp}} := (P \times P)/\!\sim$, and the addition descends from $P \times P$ to P^{gp}. ⧫

Example 1.14. (1) We have $\mathbb{N}^{\mathrm{gp}} = \mathbb{Z}$.

(2) We have $(\mathbb{N}^{\oplus 2})^{\mathrm{gp}} = \mathbb{Z}^{\oplus 2}$. ◊

Exercise 1.15. Show that $P \mapsto P^{\mathrm{gp}}$ is a well-defined functor and a left adjoint to Ab $\to$ CMon.

Lemma 1.16. *Let $\sigma \subset N_{\mathbb{R}}$ be a rational polyhedral cone. Then, we have an injective map $(P_\sigma)^{\mathrm{gp}} \to M$. If σ is strongly convex, then $(P_\sigma)^{\mathrm{gp}} \to M$ is an isomorphism.*

Proof. First, we construct a homomorphism $(P_\sigma)^{\mathrm{gp}} \to M$. We start with $P_\sigma \times P_\sigma \to M$ given by $(p, d) \mapsto p - d$. It is easy to see that this descends to the quotient $(P_\sigma)^{\mathrm{gp}} = (P \times P)/\!\sim$.

We show that $(P_\sigma)^{\mathrm{gp}} \to M$ is injective. Assume that $[p, d] \mapsto 0$. This implies $p = d$ since M is a group. But then, we have $p + 0 = d + 0$, so $[p, d] = [0, 0]$ in $(P_\sigma \times P_\sigma)/\!\sim$.

We show surjectivity under the assumption that σ is strongly convex. Let $m_1, \ldots, m_\ell \in \sigma^\vee \cap M$ be generators of the cone $\sigma^\vee$, and let $u = \sum_{j=1}^{\ell} m_j$. Let $n_1, \ldots, n_k \in \sigma \cap N$ be generators of the cone σ. For every i, we have $\langle u_j, n_i\rangle \geq 0$, so $\langle u, n_i\rangle \geq 0$. Now, suppose that $\langle u, n_i\rangle = 0$. Then, we have $\langle m_j, n_i\rangle = 0$ for all $1 \leq j \leq \ell$. But this implies $\langle m, n_i\rangle = 0$ for all $m \in \sigma^\vee$, and hence $-n_i \in \sigma$. Since σ is strongly convex, this implies $n_i = 0$, a contradiction. In particular, we have $\langle u, n_i\rangle > 0$ for all $1 \leq i \leq k$. Now, let $m \in M$, and let $\mu_i = \langle m, n_i\rangle$. Then, we have $m' = m - \min\{\mu_i, 0\}u \in \sigma^\vee$ so that $m \in (P_\sigma)^{\mathrm{gp}}$. □

Definition 1.17. A monoid P is *integral* if the following holds: Let $p_1 + p = p_2 + p$. Then, we have $p_1 = p_2$. ⧫

Lemma 1.18. *A monoid P is integral if and only if the natural map $P \to P^{\mathrm{gp}}$ given by $p \mapsto (p, 0)$ is injective.*

Proof. First, assume that the map is injective. We have $(p_1, 0) \sim (p_2, 0)$ since $p_1 + p = p_2 + p$, so injectivity yields $p_1 = p_2$. Conversely, assume that P is integral, and let $(p_1, 0) \sim (p_2, 0)$. Then, we have $p \in P$ with $p_1 + p = p_2 + p$, and integrality yields $p_1 = p_2$. □

Note that P_σ is integral. A monoid is *fine* if it is finitely generated and integral.

A monoid is *saturated* if it is integral, and if $P \to P^{\mathrm{gp}}$ has the following property: Let $m \in P^{\mathrm{gp}}$, let $k \geq 1$, and assume that $km = m + \ldots + m \in P \subseteq P^{\mathrm{gp}}$. Then, we have $m \in P$.

Example 1.19. The submonoid $\mathbb{N} \setminus \{1\} = \{0, 2, 3, 4, \ldots\} \subseteq \mathbb{N}$ is not saturated. ◊

Lemma 1.20. *Let $\sigma \subset N_{\mathbb{R}}$ be a rational polyhedral cone. Then, P_σ is saturated.*

Proof. Let $m \in M$ with $km \in P_\sigma$. Then, we have $km \in \sigma^\vee$, and hence $m \in \sigma^\vee$ because $\langle km, n\rangle \geq 0$ if and only if $\langle m, n\rangle \geq 0$ for any $n \in N_{\mathbb{R}}$. But then, we have $m \in \sigma^\vee \cap M = P_\sigma$. □

A monoid P is called *toric* if it is fine, saturated, and $P^{\mathrm{gp}} \cong \mathbb{Z}^{\oplus r}$ for some $r \geq 0$. Apparently, P_σ is a toric monoid. The converse is true as well.

Exercise 1.21. Let P be a toric monoid. Then, there is some strongly convex rational polyhedral cone σ with $P \cong P_\sigma$.

2 Affine toric varieties

Let $P = P_\sigma$ be a toric monoid. Then, we define the *monoid algebra*

$$\mathbb{C}[P] = \bigoplus_{p \in P} \mathbb{C} \cdot \mathrm{z}^p$$

via the multiplication $\mathrm{z}^{p_1} \cdot \mathrm{z}^{p_2} = \mathrm{z}^{p_1+p_2}$. The associated *affine toric variety* is

$$A_P = \operatorname{Spec} \mathbb{C}[P].$$

Example 2.1. (1) Let $P = \mathbb{N}$. Then, we have $\mathbb{C}[\mathbb{N}] \cong \mathbb{C}[t]$ and $A_{\mathbb{N}} \cong \mathbb{A}^1$.

(2) Let $P = \mathbb{N}^{\oplus 2}$. Then, we have $\mathbb{C}[P] \cong \mathbb{C}[x, y]$ and $A_P \cong \mathbb{A}^2$.

(3) Let $P = P_2 = \sigma_c^\vee \cap M$. Then, we have $\mathbb{C}[P] \cong \mathbb{C}[x, y, z]/(xy - z^2)$, and A_P has an isolated singularity in 0. ◊

Fact 2.2. *Let P be a toric monoid.*

(1) *$\mathbb{C}[P]$ is finitely generated as a $\mathbb{C}$-algebra.*

(2) *$\mathbb{C}[P]$ is an integral domain.*

(3) *$\mathbb{C}[P]$ is normal.*

(4) *The Krull dimension of $\mathbb{C}[P]$ is $\mathrm{rk}(P^{\mathrm{gp}})$.*

(5) *$\mathbb{C}[P]$ is Cohen–Macaulay.*

For any commutative monoid P, a *face* $F \subseteq P$ is a non-empty subset such that $x+y \in F$ if and only if $x \in F$ and $y \in F$.

Example 2.3. (1) The faces of $\mathbb{N}$ are $\{0\}$ and $\mathbb{N}$.

(2) The faces of $\mathbb{N}^{\oplus 2}$ are 0, $0 \oplus \mathbb{N}$, $\mathbb{N} \oplus 0$, and $\mathbb{N} \oplus \mathbb{N}$.

(3) The faces of P_2 are 0, $\mathbb{N} \cdot (0, 1)$, $\mathbb{N} \cdot (-1, 2)$, and P_2. ◊

When $F \subseteq P$ is a face, we define a closed subset $V_{P,F} \subset A_P$ via the surjection

$$\iota_{P,F}^\sharp \colon \ \mathbb{C}[P] \to \mathbb{C}[F]$$

given by $\mathrm{z}^p \mapsto \mathrm{z}^p$ if $p \in F$ and $\mathrm{z}^p \mapsto 0$ if $p \notin F$. In particular, $V_{P,F} \cong \operatorname{Spec} \mathbb{C}[F] = A_F$ as schemes.

Exercise 2.4. Show that $\iota^\sharp_{P,F}\colon \mathbb{C}[P] \to \mathbb{C}[F]$ is a well-defined ring homomorphism.

We have the *localization* $\lambda_F\colon P \to P_F$. Here, $P_F = (P \times F)/\sim$ for $(p_1, f_1) \sim (p_2, f_2)$ if there is some $f \in F$ with $p_1 + f_2 + f = p_2 + f_1 + f$. For a toric monoid $P = P_\sigma$, we also have $P_F = P - F \subseteq M$.

Exercise 2.5. Show that $\varepsilon^\sharp_{P,F}\colon \mathbb{C}[P] \to \mathbb{C}[P_F]$ induced by $\lambda_F\colon P \to P_F$ is the localization in the multiplicative system $\mathrm{z}^F = \{\mathrm{z}^f \mid f \in F\}$.

The localization map $\mathbb{C}[P] \to \mathbb{C}[P_F]$ induces an open immersion $U_{P,F} \subseteq A_P$. In the case $F = P$, we write $A^\circ_P \coloneqq U_{P,P} \cong \operatorname{Spec}\mathbb{C}[P^{\mathrm{gp}}] \cong (\mathbb{A}^1 \setminus \{0\})^r$. This is an algebraic torus.

We write $V^\circ_{P,F} \coloneqq V_{P,F} \cap U_{P,F}$ for the intersection.

Exercise 2.6. Show that $V^\circ_{P,F} \cong \operatorname{Spec}\mathbb{C}[F^{\mathrm{gp}}] \cong (\mathbb{A}^1 \setminus \{0\})^{\mathrm{rk}(F^{\mathrm{gp}})}$ is an algebraic torus.

Lemma 2.7. *Let P be a toric monoid. Then, we have a disjoint decomposition*

$$A_P = \bigsqcup_{F \subseteq P} V^\circ_{P,F}.$$

Proof. Let $\mathfrak{p} \subset \mathbb{C}[P]$ be a prime ideal, and let $F_\mathfrak{p} = \{p \in P \mid \mathrm{z}^p \notin \mathfrak{p}\}$. Then, $F_\mathfrak{p} \subseteq P$ is a face. We have a factorization $\mathbb{C}[P] \to \mathbb{C}[F] \to \mathbb{C}[P]/\mathfrak{p}$ so that $\mathfrak{p} \in V_{P,F}$. Furthermore, $\mathrm{z}^p(\mathfrak{p}) \neq 0 \in \kappa(\mathfrak{p})$, so $\mathfrak{p} \in U_{P,F}$.

To show that the decomposition is disjoint, assume that $\mathfrak{p} \in V^\circ_{P,F}$. Then, we have $\mathfrak{p} \in V_{P,F}$ and hence a factorization $\mathbb{C}[P] \to \mathbb{C}[F] \to R/\mathfrak{p}$. This shows that $F_\mathfrak{p} \subseteq F$. Furthermore, we have $\mathfrak{p} \in U_{P,F}$ so that $\mathrm{z}^p(\mathfrak{p}) \neq 0$ for all $p \in F$. But then, $\mathrm{z}^p \notin F$, and we have $F \subseteq F_\mathfrak{p}$. Thus, $F = F_\mathfrak{p}$ is uniquely determined by $\mathfrak{p}$, and the decomposition is disjoint. □

Example 2.8. For $P = \mathbb{N}$, we have $V^\circ_{\mathbb{N},0} = \{0\}$ and $V^\circ_{\mathbb{N},\mathbb{N}} = \mathbb{A}^1 \setminus \{0\}$. ◊

3 Log structures

When $\mathcal{P}$ is a sheaf of monoids on a topological space X, then $W \mapsto \Gamma(W, \mathcal{P})^\times$ is a sheaf, which we denote by $\mathcal{P}^\times$. Indeed, $(-)^\times$ is the right adjoint to $\mathrm{Ab} \to \mathrm{CMon}$, so it preserves limits.

Definition 3.1. Let X be a scheme.

(1) A *pre-log structure* is a sheaf of commutative monoids $\mathcal{M}_X$ together with a monoid homomorphism $\alpha_X\colon \mathcal{M}_X \to (\mathcal{O}_X, \cdot)$.

(2) A *log structure* is a pre-log structure $\alpha_X\colon \mathcal{M}_X \to (\mathcal{O}_X, \cdot)$ such that $\alpha_X^{-1}(\mathcal{O}_X^\times) \to \mathcal{O}_X^\times$ is an isomorphism.

(3) A *(pre-)log scheme* $X^\dagger = (X, \mathcal{M}_X)$ is a scheme with a (pre-)log structure $\alpha_X\colon \mathcal{M}_X \to (\mathcal{O}_X, \cdot)$.

(4) A *morphism* between (pre-)log schemes $(X, \mathcal{M}_X)$ and $(Y, \mathcal{M}_Y)$ consists of a morphism of schemes $f\colon X \to Y$ and a homomorphism of sheaves of monoids $f^\flat\colon \mathcal{M}_Y \to f_*\mathcal{M}_X$ such that the diagram

$$\begin{array}{ccc} \mathcal{M}_Y & \xrightarrow{\alpha_Y} & \mathcal{O}_Y \\ \downarrow{\scriptstyle f^\flat} & & \downarrow{\scriptstyle f^\sharp} \\ f_*\mathcal{M}_X & \xrightarrow{f_*\alpha_X} & f_*\mathcal{O}_X \end{array}$$

commutes.

We write $\mathrm{Sch}^{\mathrm{plog}}$ for the category of pre-log schemes, and $\mathrm{Sch}^{\log}$ for its full subcategory of log schemes. We say that X is the *underlying scheme* of the log scheme $(X, \mathcal{M}_X)$. This constitutes a (forgetful) functor $\mathrm{Sch}^{\log} \to \mathrm{Sch}$. ⧫

Remark 3.2. One usually considers log schemes where $\mathcal{M}_X$ is a sheaf in the étale topology. The theory is largely analogous to the simpler case of the Zariski topology, which we consider in this lecture. ◊

Example 3.3. (1) Let X be a scheme. Then, $\mathcal{M}_X = \mathcal{O}_X^\times$ is the *trivial log structure*.

(2) Let X be a scheme. Then, $\mathcal{M}_X = \mathcal{O}_X$ is the *empty log structure*.

(3) Let X be a scheme and $D \subset X$ be a closed subscheme. Then,

$$\Gamma(W, \mathcal{M}_{(X|D)}) = \{r \in \Gamma(W, \mathcal{O}_X) \mid r|_{W \setminus D} \text{ is invertible}\}$$

for $W \subseteq X$ is a log structure, the *divisorial log structure* or *compactifying log structure*. We write $(X|D) = (X, \mathcal{M}_{(X|D)})$ for the resulting log scheme.

(4) Let $X = \mathbb{A}^1_t$ and $D = \{0\}$. Then,

$$\Gamma(X, \mathcal{M}_{(X|D)}) = \{r \in \mathbb{C}[t] \mid r|_{\mathbb{A}^1 \setminus \{0\}} \text{ is invertible}\} = \mathbb{C}^\times \cdot \{t^m \mid m \geq 0\}$$

for the divisorial log structure. ◊

Exercise 3.4. Show that the pre-log structure $\mathcal{M}_{(X|D)} \to \mathcal{O}_X$ is indeed a log structure.

Lemma 3.5. *Let $(X, \mathcal{M}_X)$ be a log scheme. Then, we have $\mathcal{M}_X^\times = \alpha_X^{-1}(\mathcal{O}_X^\times)$.*

Proof. First, assume that $m \in \mathcal{M}_X^\times$. Then, we have $\alpha_X(m) \in \mathcal{O}_X^\times$ because $\alpha_X(-m) = \alpha_X(m)^{-1}$. Conversely, let $m \in \mathcal{M}_X$, and assume that $\alpha_X(m) \in \mathcal{O}_X^\times$. Then, there is some $u \in \mathcal{O}_X^\times$ with $u \cdot \alpha_X(m) = 1$, and we have some $n \in \mathcal{M}_X$ with $\alpha_X(n) = u$. But then, $\alpha_X(m+n) = 1$, so we have $m + n = 0$ and $m \in \mathcal{M}_X^\times$. □

Example 3.6. Let $P = P_\sigma$ be a toric monoid. Then, the map $P \to \mathbb{C}[P]$ given by $p \mapsto \mathrm{z}^p$ induces a pre-log structure $\beta\colon \underline{P}_{A_P} \to \mathcal{O}_{A_P}$ on $A_P = \operatorname{Spec} \mathbb{C}[P]$. Here, $\underline{P}_{A_P}$ is the constant sheaf of monoids with stalk P, i.e., the sheafification of $W \mapsto P$. The pre-log structure $\beta\colon \underline{P}_{A_P} \to \mathcal{O}_{A_P}$ is never a log structure. Indeed, $P^\times = (\underline{P}_{X,x})^\times$ is countable while $\mathcal{O}_{X,x}^\times$ contains $\mathbb{C}^\times$. ◊

Example 3.7. Let $P = P_\sigma$ be a toric monoid, and let $D_P = A_P \setminus A_P^\circ$ be the *toric boundary*. We write $A_P^\dagger \coloneqq (A_P|D_P)$. In a sense, this is the most fundamental log scheme of them all; we will come back to it repeatedly. We have a morphism of pre-log schemes $A_P^\dagger \to (A_P, \underline{P}_{A_P})$. ◊

4 Integrality and saturatedness

Definition 4.1. A log scheme $(X, \mathcal{M}_X)$ is *m-integral* if $\Gamma(W, \mathcal{M}_X)$ is an integral monoid for every $W \subseteq X$. We write $\mathrm{Sch}^{\log}_{\mathrm{int}}$ for the full subcategory of $\mathrm{Sch}^{\log}$ whose objects are m-integral log schemes.

A log scheme $(X, \mathcal{M}_X)$ is *saturated* if $\Gamma(W, \mathcal{M}_X)$ is an integral and saturated monoid for every $W \subseteq X$. We write $\mathrm{Sch}^{\log}_{\mathrm{sat}}$ for the full subcategory of $\mathrm{Sch}^{\log}$ whose objects are saturated log schemes. ⧫

Remark 4.2. The standard terminology is "integral," but this is potentially misleading as an integral log scheme does not need to have an integral underlying scheme. ◊

Example 4.3. (1) $(X, \mathcal{O}_X^\times)$ is always m-integral and saturated

(2) Let X be an integral scheme, and $D \subset X$ be a proper closed subset. Then, $(X|D)$ is an m-integral log scheme. In particular, $A_P^\dagger = (A_P|D_P)$ is m-integral.

(3) $A_P^\dagger$ is saturated, but this is not obvious.

(4) Let $X = \operatorname{Spec} \mathbb{C}[x,y]/(y^2 - x^3)$, let $D = \{0\}$, and consider $\frac{y}{x} \in \Gamma(X, \mathcal{K}_X)$. Then, we have $(\frac{y}{x})^2 = x \in \Gamma(X, \mathcal{M}_X) \subseteq \Gamma(X, \mathcal{O}_X)$, but $\frac{y}{x} \notin \Gamma(X, \mathcal{M}_X)$ under the embedding $\mathcal{M}_X^{\mathrm{gp}} \to \mathcal{K}_X$. Thus, $(X|D)$ is not saturated. ◊

Exercise 4.4. A log scheme $(X, \mathcal{M}_X)$ is m-integral if and only if the stalk $\mathcal{M}_{X,x}$ is an integral monoid for every point $x \to X$.